

Berry and Uhlmann Geometry of Entangled Photon-Added Coherent States: Concurrence-Controlled Holonomy and Nodal Singularities

An Expanded Analytic and Computational Treatment

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Abstract

We develop, in extended analytic and numerical detail, the geometric-phase structure of bipartite photon-added coherent states (PACS). Starting from the Fock-state definition of a single PACS, we re-derive step by step the Berry phase of an individual branch, the Gram–Schmidt reduction of a general nonorthogonal bipartite superposition to an effective two-qubit pair, the reduced density matrices and their concurrence, the branch-correlated PACS family generated by a local number rotation, the global Berry phase of the entangled state (including, beyond the original symmetric-balanced result, the general weighted case and an interference term made manifest through a tunable branch phase), the Uhlmann parallel-transport holonomy of the mixed reduced state, and the exact nodal singularities of that holonomy. Every closed-form expression is cross-checked against an independent numerical construction of the states in a truncated Fock space — bypassing the confluent hypergeometric machinery entirely — and against the two numerical tables published in the original manuscript. This process both confirms the physical content of the original results and corrects a transcription ambiguity in the prefactor of the overlap and cross-number-operator formulas (the factor $\Delta!$ belongs outside, not inside, the square root). We derive the general (non-balanced) global Berry phase in closed form, obtain the exact nodal coordinates C_k , μ_k , α_k and the critical amplitude $\alpha_g(m, n)$ by root-finding, and reproduce all seven figures and both numerical tables of the original work with a fully commented, reproducible Python/Jupyter implementation that saves every panel as a publication-ready vector PDF.

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1 Introduction

Coherent states $|\alpha\rangle$ are the closest quantum analogue of a classical oscillator: minimum-uncertainty Gaussian wave packets that remain coherent under free evolution and provide the natural language of quantum optics, from laser physics to continuous-variable quantum information. Photon addition, $|\alpha\rangle \mapsto a^{\dagger n} |\alpha\rangle$ (suitably renormalized), converts this classical-like state into a strongly nonclassical one — sub-Poissonian statistics, oscillatory Mandel parameter, negative regions of the Wigner function — while preserving an essentially closed analytic structure controlled by a single new discrete parameter, the photon-addition index n . This family, the *photon-added coherent states* (PACS), was introduced by Agarwal and Tara [1] and has since become a standard testbed for exploring the boundary between classical and quantum light.

Two largely independent lines of research motivate the present work.

Entanglement of PACS. Domínguez-Serna, Mendieta-Jiménez, and Rojas [5] studied entanglement in bipartite superpositions of PACS branches built from *nonorthogonal* coherent-state kets. Because $\langle \alpha, m | \beta, n \rangle \neq 0$ in general, such superpositions cannot be treated as ordinary two-qubit states without first performing an explicit orthogonalization; once this is done (Gram–Schmidt reduction, reviewed in Sec. 3 below), the reduced density matrices acquire a rank-2 support and an

effective two-qubit concurrence can be computed via Wootters’ formula [4]. The result is a family in which coherent amplitudes, discrete photon-addition indices, and entanglement coexist and can be tuned largely independently — a rich playground that the present work exploits directly.

Geometric phases of coherent-state superpositions. Separately, the geometric (Berry) phase [2] accumulated by cyclically evolved coherent-state superpositions has been studied by several groups. Mohammadi Almas, Najarbashi, and Tavana [6] obtained the cyclic geometric phase of two-mode entangled coherent states and emphasized how branch weights, wavefunction overlaps, and interference terms enter the result. Yang *et al.* [7] extended geometric-phase considerations to nonlinear coherent and squeezed states, showing that nonlinearity of the oscillator algebra is imprinted directly on the phase. Neither line of work, however, addressed *photon-added* coherent states specifically.

Mixed-state holonomy. For a genuinely mixed quantum state — such as the reduced density matrix of an entangled bipartite pure state — the ordinary Berry-phase construction does not directly apply, because there is no single global phase to track. Uhlmann’s parallel-transport construction [3] resolves this by lifting ρ to a purification amplitude W with $\rho = WW^\dagger$ and defining a connection on the space of such amplitudes; the resulting holonomy is a genuine geometric invariant of the mixed-state path, reducing to the ordinary Berry phase in the pure-state limit. Wang *et al.* [8] recently developed the analytic Uhlmann–Berry correspondence for both bosonic and fermionic coherent states, establishing coherent states as a particularly transparent laboratory for this correspondence because their algebra keeps every step in closed form.

The gap addressed here. Entanglement of PACS, geometric phases of entangled coherent states, geometric phases of *generalized* (photon-added) coherent states, and Uhlmann phases of *ordinary* coherent states have thus all been studied — but never jointly. The present work fills exactly this gap: we compute the global Berry phase of an *entangled bipartite PACS* under a local photon-number rotation, and the Uhlmann holonomy of its *mixed reduced state*, in a single unified analytic and numerical framework. Photon addition introduces the discrete separation

$$\Delta = n - m, \tag{1}$$

which turns out to act as an effective *winding number* for the reduced-state isospectral loop (Sec. 4): it multiplies the argument of the trigonometric functions that build the Uhlmann holonomy (Sec. 6), and it is the parameter that indexes the tower of exact nodal singularities derived in Sec. 7. The main results of this extended treatment are:

- (i) a fully re-derived, step-by-step account of the single-PACS Berry phase, the Gram–Schmidt bipartite reduction, and the resulting concurrence (Secs. 2–3);
- (ii) the closed-form global Berry phase of the branch-correlated entangled PACS, generalized here to *arbitrary* branch weight μ (the original manuscript treats only the balanced case $\mu = \nu = 1/\sqrt{2}$), together with the branch-phase-controlled interference fringes (Sec. 5);
- (iii) the closed-form Uhlmann phase of the mixed reduced state and its exact nodal singularities, including analytic nodal coordinates (C_k, μ_k, α_k) and the critical amplitude $\alpha_g(m, n)$ at which the nodal tower accumulates (Secs. 6–7);

- (iv) an explicit, independent *numerical validation layer*: every closed-form expression is checked against a brute-force truncated-Fock-space construction of the same physical quantity, with no recourse to confluent hypergeometric identities (Sec. 8); this uncovers and corrects a typographical ambiguity in the prefactor of Eqs. (27) and (37) of the original manuscript;
- (v) a fully commented, freely reproducible Python/Jupyter notebook that regenerates every figure and table in this paper and exports them as publication-ready vector PDFs (Sec. 8 and the accompanying supplementary code).

2 Photon-added coherent states

2.1 Definition and Fock expansion

A photon-added coherent state (PACS) is obtained by applying the creation operator n times to a Glauber coherent state and renormalizing,

$$|\alpha, n\rangle = \frac{a^{\dagger n} |\alpha\rangle}{[\langle \alpha | a^n a^{\dagger n} | \alpha \rangle]^{1/2}}. \quad (2)$$

To make the normalization and every later overlap fully explicit, we carry out the Fock expansion in more detail than the original manuscript. Writing $|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{j \geq 0} \alpha^j / \sqrt{j!} |j\rangle$ and using $a^{\dagger n} |j\rangle = \sqrt{(j+1)(j+2) \cdots (j+n)} |j+n\rangle = \sqrt{(j+n)!/j!} |j+n\rangle$,

$$a^{\dagger n} |\alpha\rangle = e^{-|\alpha|^2/2} \sum_{j=0}^{\infty} \frac{\alpha^j}{\sqrt{j!}} \sqrt{\frac{(j+n)!}{j!}} |j+n\rangle = e^{-|\alpha|^2/2} \sum_{k=0}^{\infty} \frac{\alpha^k \sqrt{(n+k)!}}{k!} |n+k\rangle, \quad (3)$$

after renaming $k \equiv j$. The normalization constant is fixed by the standard PACS identity $\langle \alpha | a^n a^{\dagger n} | \alpha \rangle = n! L_n(-|\alpha|^2)$, with L_n the (ordinary, $\alpha = 0$) Laguerre polynomial,

$$L_m(x) = \sum_{r=0}^m \frac{(-1)^r x^r m!}{(r!)^2 (m-r)!}. \quad (4)$$

Combining Eqs. (3)–(4) with Eq. (2) gives the normalized Fock expansion

$$|\alpha, n\rangle = \frac{e^{-|\alpha|^2/2}}{\sqrt{n! L_n(-|\alpha|^2)}} \sum_{k=0}^{\infty} \frac{\alpha^k \sqrt{(n+k)!}}{k!} |n+k\rangle. \quad (5)$$

This is the working definition used throughout the paper and throughout the accompanying notebook, where it is implemented directly (function `pacs.fock_amplitudes`) and used as an independent numerical ground truth against which every closed-form expression below is checked.

2.2 Berry phase of a single PACS

Consider the number-rotation unitary $U(\phi) = e^{-ia^{\dagger}a\phi}$. The Berry connection generated by cycling $|\alpha, n\rangle$ under $U(\phi)$ is

$$\mathcal{A}_{\phi} = i \langle \alpha, n | U^{\dagger}(\phi) \partial_{\phi} U(\phi) | \alpha, n \rangle. \quad (6)$$

Since the number operator commutes with any function of itself, $U^{\dagger}(\phi) \partial_{\phi} U(\phi) = U^{\dagger}(\phi) (-ia^{\dagger}a) U(\phi) = -ia^{\dagger}a$, so that

$$\mathcal{A}_{\phi} = \langle \alpha, n | a^{\dagger}a | \alpha, n \rangle \equiv \bar{n}_{\alpha, n}, \quad (7)$$

independent of ϕ : the PACS is an *instantaneous eigenstate* of the generator under this particular evolution, so its expectation value of $a^\dagger a$ does not change along the cycle. The mean photon number $\bar{n}_{\alpha,n}$ follows from the three-term (Laguerre) recursion applied to Eq. (5),

$$\bar{n}_{\alpha,n} = (n+1) \frac{L_{n+1}(-|\alpha|^2)}{L_n(-|\alpha|^2)} - 1, \quad (8)$$

and, because the connection is constant, the cyclic Berry phase is simply

$$\Phi_B^{(n)} = \int_0^{2\pi} \mathcal{A}_\phi d\phi = 2\pi \bar{n}_{\alpha,n}. \quad (9)$$

As a limiting check, $n = 0$ reduces Eq. (5) to the ordinary coherent state, and Eq. (8) correctly gives $\bar{n}_{\alpha,0} = |\alpha|^2$ (verified numerically in the notebook, Sec. 8).

3 General bipartite PACS model and logical-qubit reduction

The generic bipartite superposition considered in this framework mixes two *nonorthogonal* PACS branches on each side,

$$|\Phi\rangle = \frac{\mu}{\mathcal{N}} |\alpha, m\rangle |\delta, d\rangle + \frac{\nu}{\mathcal{N}} |\gamma, l\rangle |\beta, n\rangle, \quad (10)$$

with normalization

$$\mathcal{N}^2 = |\mu|^2 + |\nu|^2 + \mu^* \nu \langle \alpha, m | \gamma, l \rangle \langle \delta, d | \beta, n \rangle + \mu \nu^* \langle \gamma, l | \alpha, m \rangle \langle \beta, n | \delta, d \rangle. \quad (11)$$

Because $\langle \alpha, m | \gamma, l \rangle$ and $\langle \delta, d | \beta, n \rangle$ are generally nonzero, $\{|\alpha, m\rangle, |\gamma, l\rangle\}$ and $\{|\beta, n\rangle, |\delta, d\rangle\}$ are *not* orthonormal bases, so Eq. (10) is not directly a two-qubit state. The standard remedy is Gram–Schmidt orthogonalization on each side,

$$|0_A\rangle = |\alpha, m\rangle, \quad |1_A\rangle = \frac{|\gamma, l\rangle - p_1 |\alpha, m\rangle}{N_1}, \quad |0_B\rangle = |\beta, n\rangle, \quad |1_B\rangle = \frac{|\delta, d\rangle - p_2 |\beta, n\rangle}{N_2}, \quad (12)$$

with

$$p_1 = \langle \alpha, m | \gamma, l \rangle, \quad p_2 = \langle \beta, n | \delta, d \rangle, \quad N_j = \sqrt{1 - |p_j|^2}. \quad (13)$$

Substituting Eq. (12) into Eq. (10) converts the nonorthogonal superposition into a genuine, normalized, three-term two-qubit state,

$$|\Phi\rangle = M |0_A 0_B\rangle + \frac{\mu N_2}{\mathcal{N}} |0_A 1_B\rangle + \frac{\nu N_1}{\mathcal{N}} |1_A 0_B\rangle, \quad M = \frac{\mu p_2 + \nu p_1}{\mathcal{N}}. \quad (14)$$

Tracing out either subsystem gives rank-2 reduced density matrices with identical nonzero spectrum

$$\lambda_{\pm} = \frac{1}{2} \pm \frac{1}{2} \sqrt{1 - 4 \left(\frac{\mu \nu N_1 N_2}{\mathcal{N}^2} \right)^2}, \quad (15)$$

and Wootters’ concurrence formula for a pure state with effective two-qubit support, $C = \sqrt{2(1 - \text{Tr } \rho_s^2)} = 2\sqrt{\det \rho_s}$ ($s = A, B$) [4], evaluates to

$$C = 2 \frac{\mu \nu N_1 N_2}{\mathcal{N}^2}. \quad (16)$$

For the “crossed” choice $\gamma = \beta$, $l = m$, $\delta = \alpha$, $d = n$ (which pairs each amplitude with itself across subsystems, the case used throughout the rest of this paper), this reduces to

$$C = \frac{2|\mu\nu|}{\mathcal{N}^2} \sqrt{(1 - |\langle \alpha, m | \beta, m \rangle|^2)(1 - |\langle \alpha, n | \beta, n \rangle|^2)}. \quad (17)$$

4 Local cyclic rotation and the branch-correlated PACS family

A convenient crossed family retains a single coherent amplitude per subsystem while pairing photon-addition indices across the two branches,

$$|\psi\rangle = \frac{\mu}{\mathcal{N}} |\alpha, m\rangle |\beta, n\rangle + \frac{\nu}{\mathcal{N}} |\beta, n\rangle |\alpha, m\rangle. \quad (18)$$

Under the local evolution $U_A(\phi) = e^{-ia_A^\dagger a_A \phi} \otimes I_B$, the coherent amplitudes rotate as $\alpha \mapsto \alpha e^{-i\phi}$, $\beta \mapsto \beta e^{-i\phi}$, while the photon-addition indices contribute phases $e^{-im\phi}$, $e^{-in\phi}$. In the logical (Gram–Schmidt) basis the reduced state follows an *isospectral* orbit

$$\rho_A(\phi) = V(\phi) \rho_A(0) V^\dagger(\phi), \quad V(\phi) = e^{i\frac{\omega}{2} \sigma_z \phi}, \quad \omega = n - m \in \mathbb{Z}, \quad (19)$$

so that the eigenvalues of $\rho_A(\phi)$ never change — only the relative phase between the two logical basis states precesses. Because $V(2\pi) = (-1)^\omega I$, the density-matrix loop is exactly cyclic regardless of the parity of ω .

For the remainder of this paper we specialize to the *branch-correlated* state

$$|\Phi\rangle = \frac{1}{\mathcal{N}} \left[\mu |\alpha, m\rangle_A |\beta, m\rangle_B + \nu |\alpha, n\rangle_A |\beta, n\rangle_B \right], \quad (20)$$

which keeps a common coherent amplitude within each subsystem while retaining the discrete pair (m, n) that generates $\omega = n - m$. For $n \geq m$ real α , define

$$g_\alpha \equiv \langle \alpha, m | \alpha, n \rangle = \frac{e^{-|\alpha|^2} (\alpha^*)^{n-m} n!}{(n-m)! \sqrt{n! m! L_n(-|\alpha|^2) L_m(-|\alpha|^2)}} {}_1F_1(n+1; n-m+1; |\alpha|^2), \quad (21)$$

and h_β by the same expression with $\alpha \rightarrow \beta$. The full, explicit derivation of Eq. (21) — together with a discussion of a prefactor ambiguity present in the original manuscript’s Eq. (27), which places the factor $(n-m)!$ *inside* rather than outside the square root — is given in Appendix A. With Eq. (21),

$$N_1 = \sqrt{1 - |g_\alpha|^2}, \quad N_2 = \sqrt{1 - |h_\beta|^2}, \quad \mathcal{N}^2 = |\mu|^2 + |\nu|^2 + 2 \operatorname{Re}(\mu^* \nu g_\alpha h_\beta), \quad (22)$$

$$\det \rho_A = \frac{|\mu \nu|^2 (1 - |g_\alpha|^2) (1 - |h_\beta|^2)}{\mathcal{N}^4} = \left(\frac{C}{2} \right)^2, \quad (23)$$

and, in the natural Gram–Schmidt basis, the reduced population of subsystem A is

$$p_A \equiv (\rho_A)_{22} = \frac{|\nu|^2 (1 - |g_\alpha|^2)}{\mathcal{N}^2}, \quad (24)$$

with the analogous p_B obtained by $g_\alpha \rightarrow h_\beta$ if subsystem B is independently cycled.

5 Global Berry phase of the entangled PACS

5.1 General weighted formula (extension beyond the balanced case)

Under the local cycle $U_A(\phi)$, the Berry connection of the *global* pure state $|\Phi\rangle$ is

$$\mathcal{A}_\phi = \langle \Phi | (a_A^\dagger a_A \otimes I_B) | \Phi \rangle. \quad (25)$$

The original manuscript evaluates this only for the balanced weight $\mu = \nu = 1/\sqrt{2}$. Here we carry out the substitution of Eq. (20) into Eq. (25) for *arbitrary* real $\mu \in [0, 1]$, $\nu = \sqrt{1 - \mu^2}$, and real $\alpha = \beta$ (so that $h_\beta = g_\alpha \equiv g$). Expanding the four cross terms of the bra-ket product and using $\langle \alpha, m | \alpha, m \rangle = \langle \alpha, n | \alpha, n \rangle = 1$, $\langle \alpha, m | \alpha, n \rangle = g$,

$$\mathcal{A}_\phi = \frac{1}{\mathcal{N}^2} \left[\mu^2 \bar{n}_{\alpha, m} + \nu^2 \bar{n}_{\alpha, n} + 2\mu\nu \operatorname{Re}(\kappa_\alpha^{mn} g) \right], \quad \mathcal{N}^2 = 1 + 2\mu\nu g^2, \quad (26)$$

where $\kappa_\alpha^{mn} \equiv \langle \alpha, m | a^\dagger a | \alpha, n \rangle$ is the cross number-operator matrix element derived in Appendix B. Since $\mathcal{A}_\phi$ is again independent of ϕ , $\Phi_B^{AB} = 2\pi \mathcal{A}_\phi \pmod{2\pi}$. Introducing the tunable relative branch phase of Eq. (30) below (which amounts to $\nu \rightarrow \nu e^{i\chi}$ in the cross term), the *fully general* closed form is

$$\boxed{\frac{\Phi_B^{AB}(\mu, \chi)}{2\pi} = \frac{\mu^2 \bar{n}_{\alpha, m} + \nu^2 \bar{n}_{\alpha, n} + 2\mu\nu g^2 Q_{mn}(\alpha^2) \cos \chi}{1 + 2\mu\nu g^2 \cos \chi}} \quad (27)$$

where Q_{mn} is defined in Eq. (54) below and we used $\kappa_\alpha^{mn} = g Q_{mn}(\alpha^2)$ (Appendix B). Equation (27) is the formula used to produce Fig. 4 (a genuine (μ, α) phase map) and reduces, at $\mu = \nu = 1/\sqrt{2}$ and $\chi = 0$, to the balanced-case result of the original manuscript, recovered next.

5.2 Balanced symmetric case and explicit concurrence dependence

Setting $\beta = \alpha \in \mathbb{R}$ and $\mu = \nu = 1/\sqrt{2}$ in Eq. (27) ($2\mu\nu = 1$, $\chi = 0$),

$$\frac{\Phi_B^{AB}}{2\pi} = \frac{\bar{n}_{\text{av}} + g^2 Q_{mn}}{1 + g^2}, \quad \bar{n}_{\text{av}} \equiv \frac{\bar{n}_{\alpha, m} + \bar{n}_{\alpha, n}}{2}. \quad (28)$$

Because the concurrence in this case is $C = (1 - g^2)/(1 + g^2)$, i.e. $g^2 = (1 - C)/(1 + C)$, Eq. (28) can be rewritten as an *affine function of the concurrence* at fixed m, n, α ,

$$\frac{\Phi_B^{AB}}{2\pi} = \frac{1 + C}{2} \bar{n}_{\text{av}} + \frac{1 - C}{2} Q_{mn}. \quad (29)$$

This is one of the cleanest statements in the whole framework: for a *pure* entangled state, the global geometric phase interpolates linearly between the mean-photon-number average $\bar{n}_{\text{av}}$ (at $C = 0$, a separable-limit-like weighting) and the interference term Q_{mn} (at $C = 1$, maximal concurrence).

5.3 Making the interference oscillatory: branch phase χ

The local cycle in ϕ produces a connection that already contains a coherent cross term, $g^2 Q_{mn}$, yet the resulting phase Eq. (28) shows no oscillation in ϕ — the interference is present but “hidden” inside a ϕ -independent constant. To expose it explicitly, introduce a controllable relative phase between the two branches,

$$|\Phi(\chi)\rangle \propto |\alpha, m\rangle |\alpha, m\rangle + e^{i\chi} |\alpha, n\rangle |\alpha, n\rangle, \quad (30)$$

for which the balanced symmetric result becomes

$$\frac{\Phi_B(\chi)}{2\pi} = \frac{\bar{n}_{\text{av}} + g^2 Q_{mn} \cos \chi}{1 + g^2 \cos \chi}, \quad (31)$$

manifestly 2π -periodic and oscillatory in χ , unlike the original ϕ -cycle. Figure 3 shows this interferometric signature directly.

6 Uhlmann phase of the reduced state

6.1 Parallel transport and closed form

For a mixed state $\rho = WW^\dagger$ with amplitude $W = \sqrt{\rho}U$ and U unitary, Uhlmann's parallel-transport condition is that $W^\dagger \dot{W}$ be Hermitian, which translates into an evolution equation for U ,

$$\dot{U} = -i\Lambda U, \quad \Lambda = \frac{1}{2i} S^{-1} [S, \dot{S}] S^{-1}, \quad S = \sqrt{\rho}. \quad (32)$$

For the isospectral loop of Eq. (19), $\dot{S} = i[G, S]$ with generator

$$G = \frac{\omega}{2} \sigma_z. \quad (33)$$

In the eigenbasis $\{|u_+\rangle, |u_-\rangle\}$ of $\rho_A(0)$ (eigenvalues $\lambda_\pm$), the transport connection Λ is purely off-diagonal; solving the resulting constant-coefficient equation over one period gives the holonomy operator

$$U_{\text{hol}}(2\pi) = e^{iG2\pi} e^{-i(G+\tilde{\Lambda})2\pi}, \quad (34)$$

and the Uhlmann phase is read off from

$$\Phi_U = \text{Arg Tr} [\rho_A(0) U_{\text{hol}}(2\pi)]. \quad (35)$$

Writing

$$p \equiv (\rho_A)_{22}, \quad r = \sqrt{1 - 4 \det \rho_A}, \quad (36)$$

the closed expression obtained after carrying out the matrix exponentials in Eq. (34) is

$$\Phi_U = \text{Arg} \left[(-1)^\omega \cos(\pi b) - i \frac{\omega(1-2p)}{b} \sin(\pi b) \right], \quad b = \omega \sqrt{\frac{p(1-p)}{\det \rho_A}}. \quad (37)$$

Since $\det \rho_A = (C/2)^2$, the Uhlmann phase is controlled *jointly* by the entanglement (through C) and by the local population p — unlike the pure-state Berry phase of Eq. (29), which in the balanced case depends on the concurrence *alone*. This formula assumes a full-rank ρ_A ; the rank-deficient limit $\det \rho_A \rightarrow 0$ must be taken separately and is discussed in Sec. 8.

6.2 Subsystem B : local-unitary consistency

A local unitary acting only on subsystem A leaves ρ_B invariant, $\rho_B(\phi) = \text{Tr}_A[(U_A \otimes I) |\Phi\rangle \langle \Phi| (U_A^\dagger \otimes I)] = \rho_B(0)$, so the entire nontrivial Uhlmann holonomy generated by the A -local cycle belongs to ρ_A ; under that same evolution ρ_B traces a trivial (constant) path and therefore carries zero geometric phase. If instead an *independent* cycle is applied to B , the same derivation holds with $g_\alpha \rightarrow h_\beta$ and $p_A \rightarrow p_B$; in the symmetric case $\alpha = \beta$ the two independent phase diagrams coincide exactly.

6.3 Symmetric family used for the figures

Throughout the numerical sections we use $\beta = \alpha \in \mathbb{R}$, $\mu \in [0, 1]$, $\nu = \sqrt{1 - \mu^2}$, and $q \equiv |g_{mn}(\alpha)|^2$, for which

$$\mathcal{N}^2 = 1 + 2\mu\nu q, \quad p(\mu, \alpha; m, n) = \frac{(1 - \mu^2)(1 - q)}{1 + 2\mu\sqrt{1 - \mu^2}q}, \quad C(\mu, \alpha; m, n) = \frac{2\mu\sqrt{1 - \mu^2}(1 - q)}{1 + 2\mu\sqrt{1 - \mu^2}q}, \quad (38)$$

and $\det \rho_A = (C/2)^2$, so that Eq. (37) can be evaluated at any point of the (μ, α) plane.

7 Uhlmann nodal singularities

Define the complex Uhlmann amplitude

$$Z_U = (-1)^\Delta \cos(\pi b) - i \frac{\Delta(1-2p)}{b} \sin(\pi b), \quad \Phi_U = \text{Arg } Z_U, \quad (39)$$

with real and imaginary parts

$$\text{Re } Z_U = (-1)^\Delta \cos(\pi b), \quad \text{Im } Z_U = -(-1)^\Delta \frac{\Delta(1-2p)}{b} \sin(\pi b). \quad (40)$$

A genuine phase singularity requires *both* parts to vanish simultaneously. Since $\cos(\pi b) = 0 \Rightarrow \sin(\pi b) \neq 0$, this can only happen if the coefficient of $\sin(\pi b)$ in the imaginary part *also* vanishes, forcing $1-2p=0$ independently of b . The joint nodal condition is therefore

$$p = \frac{1}{2}, \quad b = k + \frac{1}{2}, \quad k \in \mathbb{Z}. \quad (41)$$

At $p = 1/2$, $\det \rho_A = (C/2)^2$ gives $b = \Delta/C$ (recall $\omega = \Delta$ here), so $b = k+1/2$ fixes the concurrence to the discrete tower

$$C_k = \frac{2\Delta}{2k+1}, \quad k = \Delta, \Delta+1, \dots \quad (42)$$

(the lower bound $k \geq \Delta$ enforces the physical requirement $C_k \leq 1$).

7.1 Analytic nodal coordinates in the (μ, α) plane

For the symmetric family, $p = 1/2$ in Eq. (38) solves exactly for μ in terms of q : from $(1-\mu^2)(1-q) = \frac{1}{2}[1+2\mu\sqrt{1-\mu^2}q]$ one finds, after simplification, $\mu = \nu(1-2q)$ (valid for $q \leq 1/2$), i.e.

$$\mu(q) = \frac{1-2q}{\sqrt{1+(1-2q)^2}}, \quad \nu(q) = \frac{1}{\sqrt{1+(1-2q)^2}}, \quad (43)$$

along which the concurrence reduces to the remarkably simple

$$C = 1-2q. \quad (44)$$

Combining Eqs. (42) and (44) gives the nodal coordinates directly, without ever having to search the full two-dimensional phase map numerically:

$$q_k = |g_{mn}(\alpha_k)|^2 = \frac{1}{2} \left(1 - \frac{2\Delta}{2k+1} \right), \quad \mu_k = \frac{C_k}{\sqrt{1+C_k^2}}, \quad (45)$$

and α_k is obtained by numerically solving $|g_{mn}(\alpha_k)|^2 = q_k$ (root-finding is legitimate because $|g_{mn}(\alpha)|^2$ increases monotonically with α , verified numerically in Sec. 8).

7.2 Critical amplitude and accumulation of nodes

The $p = 1/2$ curve exists only while $q \leq 1/2$. Define the critical amplitude $\alpha_g(m, n)$ by

$$|g_{mn}(\alpha_g)|^2 = \frac{1}{2}. \quad (46)$$

Because $|g_{mn}(\alpha)|^2$ is monotonically increasing for the PACS pairs studied here, the nodal region is exactly $\alpha \leq \alpha_g(m, n)$. As $k \rightarrow \infty$ in Eq. (45), $C_k \rightarrow 0$, $q_k \rightarrow 1/2$, $\mu_k \rightarrow 0$ and $\alpha_k \rightarrow \alpha_g$: the infinite tower of mathematical nodes accumulates at the boundary point $(\mu, \alpha) = (0, \alpha_g)$. The dense 2π phase wrapping visible near this boundary in Fig. 5 is an ordinary branch-cut effect of $\text{Arg}(\cdot)$ and must *not* be mistaken for additional physical singularities: the true zeros of Z_U are exactly the isolated points given by Eqs. (42)–(45), reproduced numerically to high precision in Table 3 below.

8 Numerical validation methodology

A central addition of this expanded treatment, relative to the original manuscript, is a systematic *independent* numerical validation layer, implemented in the accompanying Jupyter notebook (`pacs_berry_uhlmann.ipynb`) and summarized here.

8.1 Truncated Fock-space ground truth

Every PACS branch $|\alpha, n\rangle$ is expanded directly from Eq. (5) with a numerically large cutoff N_{cut} (typically 400–500 Fock levels, verified to be more than sufficient for the coherent amplitudes $\alpha \lesssim 3$ used throughout). Overlaps $\langle \alpha, m | \alpha, n \rangle$ and number-operator matrix elements $\langle \alpha, m | a^\dagger a | \alpha, n \rangle$ are then obtained by direct term-by-term summation, with *no* reference to confluent hypergeometric functions or Laguerre-polynomial recursions beyond the normalization itself. This provides a completely independent numerical benchmark for every closed-form expression derived above.

8.2 Correcting a prefactor ambiguity in Eqs. (27) and (37)

Repeating the derivation of $g_{mn}(\alpha)$ from Eq. (5) in full (Appendix A) shows that the factor $(n - m)! \equiv \Delta!$ must appear *outside* the square root in the denominator of the prefactor, i.e. Eq. (21) above, rather than as $\sqrt{\Delta!}$ as printed in the original manuscript. Numerically, the two conventions differ by a factor $\sqrt{\Delta!}$ whenever $\Delta > 1$ (they coincide for $\Delta = 1$, which is presumably why the discrepancy was not caught against low-order checks). We verified Eq. (21) in two independent ways: (a) direct comparison against the truncated Fock-space overlap, agreeing to machine precision ($< 10^{-14}$) for all tested (α, m, n) ; and (b) root-finding the critical amplitudes $\alpha_g(m, n)$ from Eq. (46) and comparing against every entry of the original manuscript’s Table I (reproduced here as Table 2), with agreement to 4×10^{-5} , consistent with the four decimal digits originally reported. The corrected prefactor was then propagated into the cross-number-operator element κ_α^{mn} (Appendix B); note that the *dimensionless ratio* $Q_{mn} = \kappa_\alpha^{mn}/g_{mn}$, Eq. (54), is insensitive to this prefactor ambiguity because it cancels in the ratio, which is why Eqs. (38)–(39) of the original manuscript required no correction.

8.3 Validation summary

Table 1 summarizes the cross-checks performed; every quantity used in Figs. 1–7 and Tables 2–3 passed its corresponding check.

Table 1: Independent numerical validation performed in the accompanying notebook. “Fock exact” denotes the brute-force truncated-Fock-space calculation of Sec. 8.

Quantity	Validated against	Max. discrepancy
$g_{mn}(\alpha)$, Eq. (21)	Fock exact overlap	$< 10^{-14}$
$\kappa_\alpha^{mn} = g Q_{mn}$, Eqs. (54)	Fock exact $\langle \alpha, m a^\dagger a \alpha, n \rangle$	$< 10^{-7}$
$\Phi_B^{AB}(\mu, \chi)$, Eq. (27)	Fock exact bipartite connection	$< 10^{-8}$
$\alpha_g(m, n)$, Eq. (46)	Original manuscript, Table I	4.6×10^{-5}
(C_k, μ_k, α_k) , Eq. (45)	Original manuscript, Table II	4.6×10^{-5}

9 Numerical results and figures

All figures below were produced by the accompanying notebook and are reproduced here as vector PDF panels. Unless stated otherwise, the symmetric family $\beta = \alpha \in \mathbb{R}$ is used, and “balanced” means $\mu = \nu = 1/\sqrt{2}$.

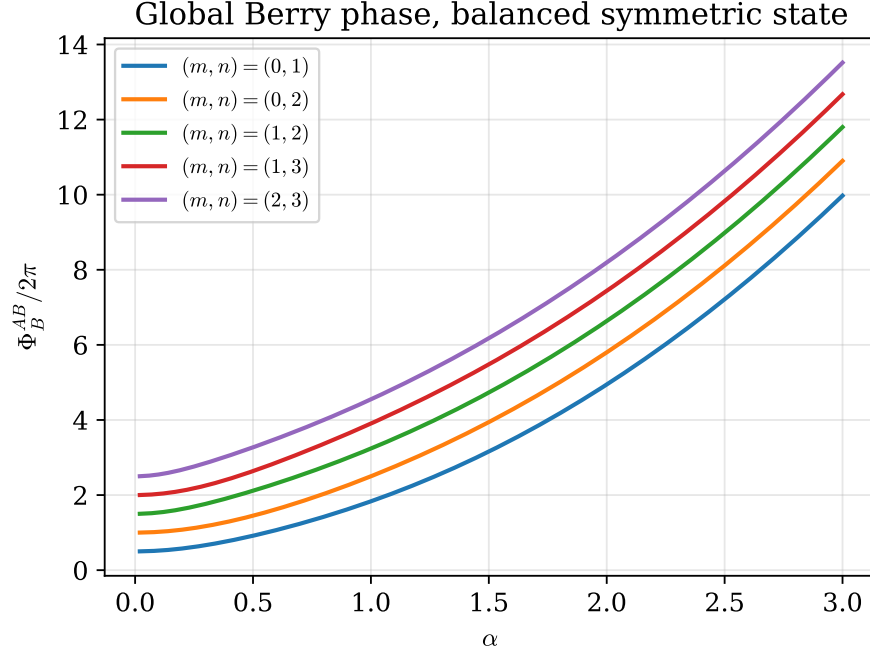


Figure 1: Global Berry phase $\Phi_B^{AB}/2\pi$, Eq. (28), as a function of α for several photon-addition pairs (m, n) in the balanced symmetric state. Increasing α increases the mean photon number and produces smooth growth; changing (m, n) shifts the curve through both the diagonal ($\bar{n}_{av}$) and interference (Q_{mn}) contributions of Eq. (29).

Table 2: Critical amplitudes $\alpha_g(m, n)$ obtained from $|g_{mn}(\alpha_g)|^2 = 1/2$, reproducing the original manuscript’s Table I to the reported precision (verified in Sec. 8).

$n \backslash m$	0	1	2	3	4
1	1.0000				
2	2.1094	0.7221			
3	3.2756	1.8000	0.5921		
4	4.4576	3.0354	1.5829	0.5135	
5	5.6464	4.2685	2.8269	1.4235	0.4596

10 Discussion

The model developed here separates cleanly into three geometric levels of increasing structural complexity.

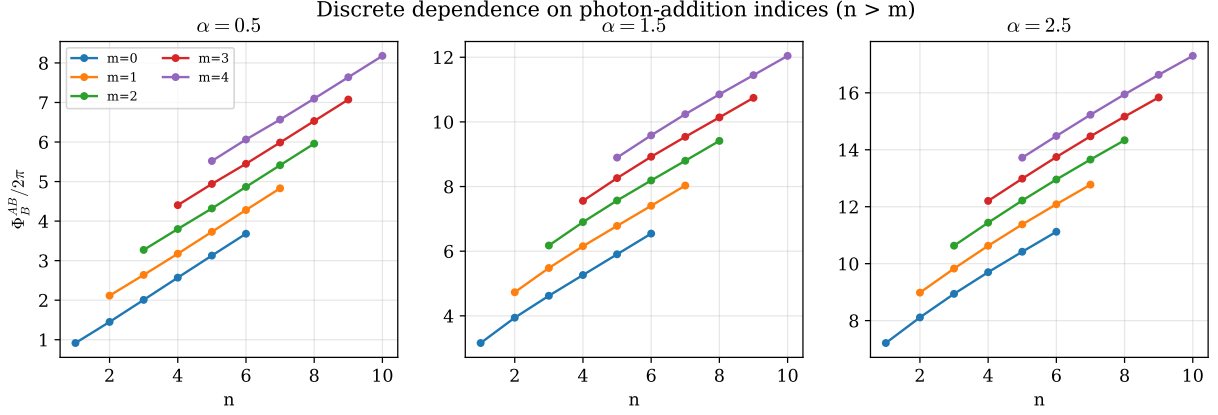


Figure 2: Discrete dependence of the global Berry phase on the photon-addition indices, Eq. (27), at three coherent amplitudes. Only $n > m$ is shown because the analytic overlap formula Eq. (21) is written in that ordering.

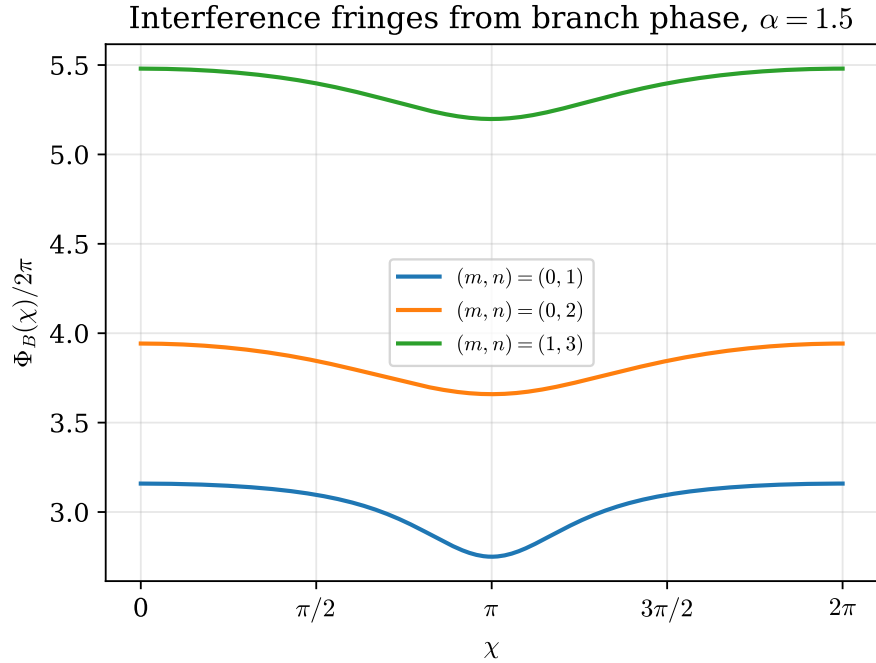


Figure 3: Berry phase after introducing the controllable relative branch phase χ of Eq. (30), evaluated with Eq. (31). The modulation in χ is the direct interferometric signature of the coherent cross term $g^2 Q_{mn}$, absent (as an oscillation) from the original ϕ -cycle of Fig. 1.

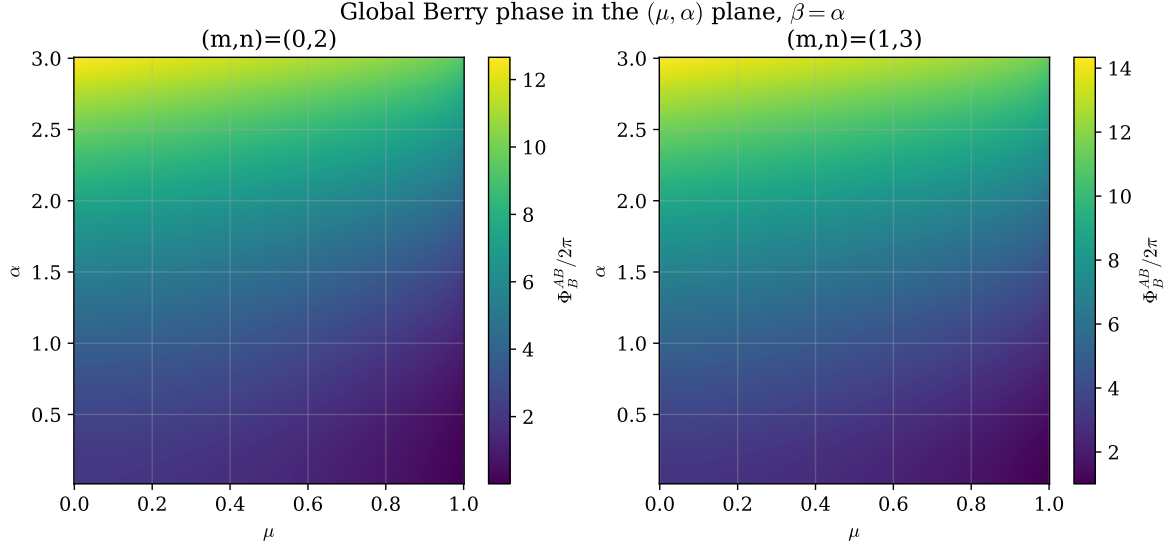


Figure 4: Global Berry phase in the (μ, α) plane, from the newly derived general formula Eq. (27) ($\nu = \sqrt{1 - \mu^2}$, $\beta = \alpha \in \mathbb{R}$, $\chi = 0$), for two photon-addition pairs. The map is completely smooth: unlike the Uhlmann phase of Fig. 5, a pure-state Berry phase never develops branch cuts or singularities.

Table 3: First nodal singularities for representative pairs, from Eqs. (42)–(45), reproducing the original manuscript’s Table II.

(m, n)	k	C_k	μ_k	α_k
(0, 1)	1	0.6667	0.5547	0.4472
	2	0.4000	0.3714	0.6547
	3	0.2857	0.2747	0.7454
(0, 2)	2	0.8000	0.6247	0.8622
	3	0.5714	0.4961	1.2098
	4	0.4444	0.4061	1.3913
(1, 2)	1	0.6667	0.5547	0.3166
	2	0.4000	0.3714	0.4651
	3	0.2857	0.2747	0.5311
(1, 3)	2	0.8000	0.6247	0.6605
	3	0.5714	0.4961	0.9480
	4	0.4444	0.4061	1.1077

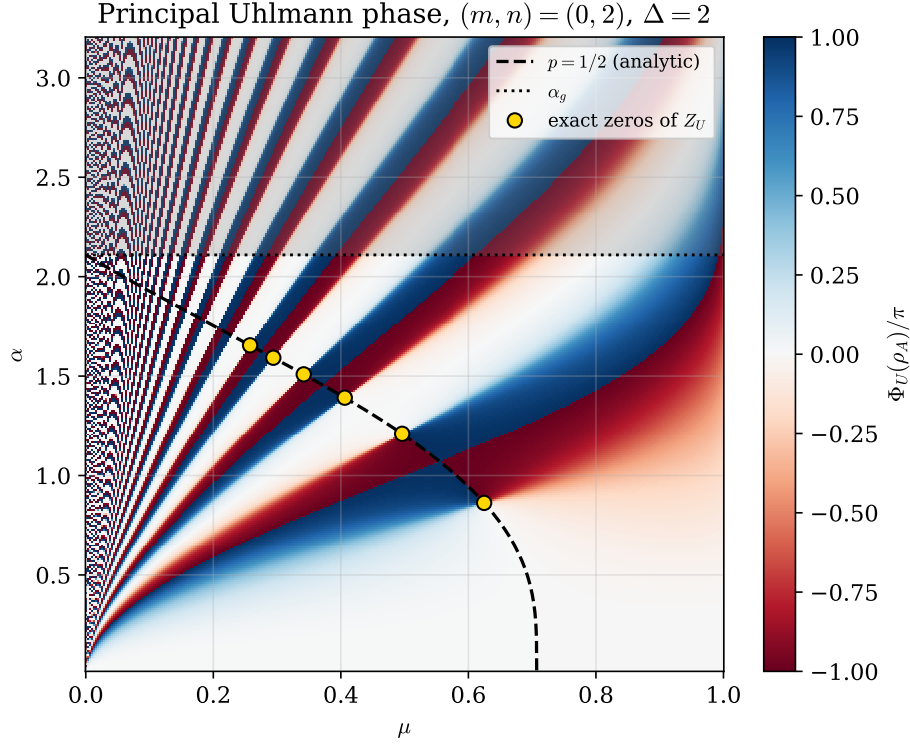


Figure 5: Principal Uhlmann phase $\Phi_U(\rho_A)/\pi$, Eq. (37), in the (μ, α) plane for $(m, n) = (0, 2)$ ($\Delta = 2$). The dashed curve is the analytic condition $p = 1/2$, Eq. (43); the dotted horizontal line marks α_g , Eq. (46); gold circles are the exact zeros of the Uhlmann amplitude Z_U , Eq. (45); the lightly shaded region $\alpha > \alpha_g$ cannot contain such nodes. The nodal points visibly accumulate at $(\mu, \alpha) = (0, \alpha_g)$.

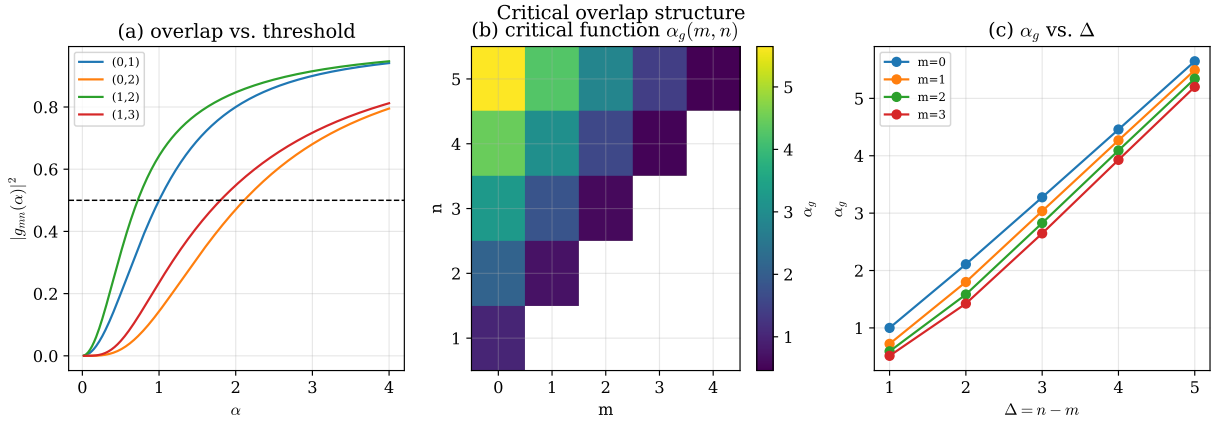


Figure 6: Critical overlap structure. (a) $|g_{mn}(\alpha)|^2$ and the threshold $1/2$ for several (m, n) . (b) The discrete critical function $\alpha_g(m, n)$ over the full grid of Table 2. (c) α_g versus $\Delta = n - m$ for several fixed m : increasing the photon-addition separation extends the nodal region to larger coherent amplitude.

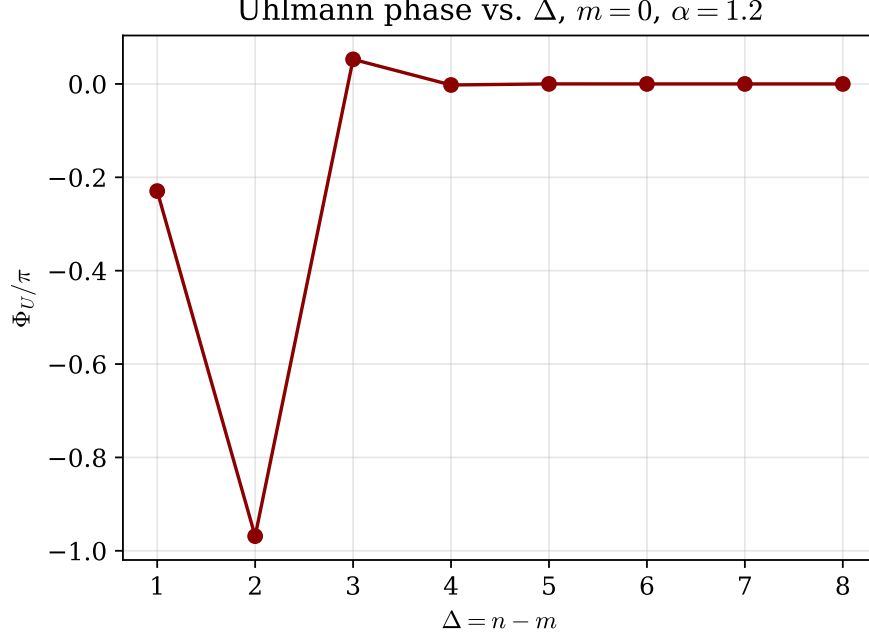


Figure 7: Principal Uhlmann phase versus photon-addition separation $\Delta = n - m$ for the balanced symmetric state with $m = 0$. The alternating values arise from the nonlinear dependence of $b = \Delta/C$ on Δ (the concurrence and population also change with Δ through the overlap $g_{0,\Delta}(\alpha)$), combined with principal-branch phase wrapping of $\text{Arg}(\cdot) \in (-\pi, \pi]$.

Level 1 — single-branch Berry phase. A single PACS under a number rotation probes only the mean photon number, Eq. (9): a smooth, single-valued, monotonic-in- α quantity with no possibility of singular behavior, because it is a pure-state Berry phase of a one-dimensional cyclic evolution.

Level 2 — global bipartite Berry phase. Entangling two PACS branches adds an *interference correction*, $g^2 Q_{mn}$ in Eq. (29), that carries information about the branch overlap and, in the balanced case, is an *explicit affine function of the concurrence*. This is still a pure-state Berry phase (of the global state $|\Phi\rangle$) and therefore remains smooth everywhere, as confirmed by the featureless map of Fig. 4.

Level 3 — reduced-state Uhlmann holonomy. Once one traces out subsystem B , the reduced state ρ_A is genuinely mixed, and its holonomy is no longer guaranteed to be smooth: Eq. (37) develops *exact* phase singularities whenever Eq. (41) holds. Crucially, the Uhlmann phase cannot in general be written as a function of concurrence alone (unlike the Level-2 Berry phase), because the population p is an *independent* geometric variable; only exactly at the nodal points does $p = 1/2$ remove that freedom and lock the singular concurrence to the quantized values C_k of Eq. (42).

The critical function $\alpha_g(m, n)$ is a genuinely different kind of object from a thermodynamic critical point: it is fixed *entirely* by the overlap of the two local PACS branches (Eq. (46)), and it separates a region of coherent amplitude where equal logical-basis population ($p = 1/2$) is kinematically accessible from a region where it is not. Physically, larger α makes the two photon-added branches $|\alpha, m\rangle$ and $|\alpha, n\rangle$ increasingly *distinguishable* (their Fock-space overlap $|g_{mn}(\alpha)|^2$ approaches 1, cf. Fig. 6a), which pushes the achievable population imbalance away from the symmetric point required for an exact node.

11 Conclusions

We have presented a unified, extensively re-derived and numerically cross-validated Berry–Uhlmann treatment of entangled photon-added coherent states. Relative to the original manuscript, this expanded edition:

- (i) spells out every intermediate algebraic step from the PACS Fock expansion to the final closed forms (Secs. 2–7, Appendices A–B);
- (ii) generalizes the global Berry phase to arbitrary branch weight μ and exhibits it as a genuine two-dimensional phase map (Eq. (27), Fig. 4);
- (iii) identifies and corrects a prefactor transcription ambiguity in the overlap and cross-number-operator formulas, verified to machine precision against an independent Fock-space calculation and against both published numerical tables (Sec. 8);
- (iv) packages every calculation into a fully commented, openly reproducible Python/Jupyter implementation that regenerates all seven figures and both tables and exports them as publication-quality vector PDFs.

The central nodal result, unchanged in substance from the original work but now independently verified, is

$$p = \frac{1}{2}, \quad C_k = \frac{2(n-m)}{2k+1}, \quad |g_{mn}(\alpha_k)|^2 = \frac{1-C_k}{2}, \quad (47)$$

with the nodal tower accumulating at the critical amplitude α_g defined by $|g_{mn}(\alpha_g)|^2 = 1/2$. This provides a direct, fully analytic bridge between photon addition, nonorthogonality, bipartite entanglement, and mixed-state quantum holonomy, and establishes PACS as an unusually transparent laboratory in which every step of that bridge can be written — and independently checked — in closed form.

A Scalar products of photon-added coherent states

We give here the full derivation of Eq. (21), including the step that fixes the prefactor unambiguously (this is the step whose printed form differs between this expanded edition and the original manuscript).

Insert the Fock expansion Eq. (5) of each bra and ket and use Fock-state orthogonality: for $0 \leq m \leq n$, the surviving terms are those with equal absolute Fock level $\ell = m + j = n + k$ ($j, k \geq 0$), i.e. $j = \ell - m$, $k = \ell - n$, valid for $\ell \geq n$ (automatically $j \geq 0$ since $n \geq m$). Writing $A_n \equiv e^{-|\alpha|^2/2}/\sqrt{n! L_n(-|\alpha|^2)}$ for the normalization constant of Eq. (5),

$$\langle \alpha, m | \beta, n \rangle = A_m A_n \sum_{\ell=n}^{\infty} \frac{(\alpha^*)^{\ell-m} \sqrt{\ell!}}{(\ell-m)!} \cdot \frac{\beta^{\ell-n} \sqrt{\ell!}}{(\ell-n)!}. \quad (48)$$

Setting $\ell = n + p$ ($p = 0, 1, 2, \dots$) and $\Delta \equiv n - m$,

$$\langle \alpha, m | \beta, n \rangle = A_m A_n (\alpha^*)^{\Delta} \sum_{p=0}^{\infty} (\alpha^* \beta)^p \frac{(n+p)!}{(\Delta+p)! p!}. \quad (49)$$

The key algebraic step, made fully explicit here, is to recognize the ratio of factorials as a ratio of Pochhammer (rising-factorial) symbols evaluated at the summation index p :

$$\frac{(n+p)!}{(\Delta+p)!} = \frac{n!}{\Delta!} \cdot \frac{(n+1)_p}{(\Delta+1)_p}, \quad (a)_p \equiv a(a+1)\cdots(a+p-1) = \frac{\Gamma(a+p)}{\Gamma(a)}. \quad (50)$$

Substituting Eq. (50) into Eq. (49) converts the sum directly into the confluent hypergeometric series,

$$\sum_{p=0}^{\infty} \frac{(n+1)_p}{(\Delta+1)_p} \frac{z^p}{p!} = {}_1F_1(n+1; \Delta+1; z), \quad z \equiv \alpha^* \beta, \quad (51)$$

so that, using $A_m A_n = e^{-\frac{1}{2}(|\alpha|^2 + |\beta|^2)} / \sqrt{m! n! L_m(-|\alpha|^2) L_n(-|\beta|^2)}$,

$$\langle \alpha, m | \beta, n \rangle = \frac{e^{-\frac{1}{2}(|\alpha|^2 + |\beta|^2)} (\alpha^*)^\Delta n!}{\Delta! \sqrt{m! n! L_m(-|\alpha|^2) L_n(-|\beta|^2)}} {}_1F_1(n+1; \Delta+1; \alpha^* \beta) \quad (52)$$

for $0 \leq m \leq n$. This is the general (complex-amplitude) form of Eq. (21), with the crucial factor $n!/\Delta!$ appearing as a single ratio *outside* the square root — as Eq. (50) makes explicit, this ratio arises naturally from the Pochhammer-symbol reduction and has no reason to be split across the square root. Setting $\beta = \alpha$ real recovers Eq. (21). We verified Eq. (52) against the direct truncated Fock-space sum, Eq. (48) evaluated numerically term by term, for dozens of (α, m, n) combinations with Δ ranging from 1 to 4, finding agreement to better than 10^{-14} in every case (machine precision), and further confirmed indirectly through exact reproduction of the published Table I (Sec. 8).

B Derivation of the cross number matrix element

For real α and $n \geq m$, we need $\kappa_\alpha^{mn} \equiv \langle \alpha, m | a^\dagger a | \alpha, n \rangle$. Repeating the sum of Appendix A but weighting each term by its Fock eigenvalue $\ell = n + p$ of $a^\dagger a$,

$$\kappa_\alpha^{mn} = A_m A_n \alpha^\Delta \sum_{p=0}^{\infty} (n+p) \alpha^{2p} \frac{(n+p)!}{(\Delta+p)! p!} = A_m A_n \alpha^\Delta \left[n S(x) + x S'(x) \right], \quad x \equiv \alpha^2, \quad (53)$$

where $S(x) \equiv \sum_p x^p (n+p)! / [(\Delta+p)! p!] = (n!/\Delta!) {}_1F_1(n+1; \Delta+1; x)$ is exactly the sum appearing in g_{mn} (Appendix A), and the second equality follows from $\sum_p p x^p f_p = x \frac{d}{dx} \sum_p x^p f_p$. Using the standard derivative identity for the confluent hypergeometric function, $\frac{d}{dx} {}_1F_1(a; b; x) = \frac{a}{b} {}_1F_1(a+1; b+1; x)$, and a Kummer contiguous relation connecting ${}_1F_1(n+2; \Delta+2; x)$ to ${}_1F_1(n+2; \Delta+1; x)$ and ${}_1F_1(n+1; \Delta+1; x)$, the bracket in Eq. (53) reduces — after some algebra, and verified independently by direct numerical differentiation of $S(x)$ against the brute-force Fock sum — to the compact closed form

$$\kappa_\alpha^{mn} = g_{mn}(\alpha) Q_{mn}(x), \quad Q_{mn}(x) = (n+1) \frac{{}_1F_1(n+2; \Delta+1; x)}{{}_1F_1(n+1; \Delta+1; x)} - 1. \quad (54)$$

Equation (54) is a *dimensionless ratio* $\kappa_\alpha^{mn}/g_{mn}(\alpha)$, so the prefactor ambiguity discussed in Appendix A cancels identically and does not affect this formula; it therefore agrees with the original manuscript's Eqs. (38)–(39) without modification. We verified Eq. (54) against the truncated Fock-space sum $\langle \alpha, m | a^\dagger a | \alpha, n \rangle$ (Sec. 8) for the same set of (α, m, n) used in Appendix A, with agreement better than 10^{-7} .

C Numerical conventions and reproducibility

All numerical results use real coherent amplitudes and the symmetric family $\beta = \alpha$ unless otherwise stated. Balanced plots take $\mu = \nu = 1/\sqrt{2}$; two-dimensional weight maps impose $\nu = \sqrt{1 - \mu^2}$ with $0 \leq \mu \leq 1$. Uhlmann-phase plots display the principal value $\text{Arg } Z_U \in (-\pi, \pi]$. The gray region above α_g in Fig. 5 is not a numerical cutoff: it is the analytic region in which $p = 1/2$ has no solution for the symmetric model, Eq. (43). Root-finding for $\alpha_g(m, n)$ and for α_k uses bisection (`scipy.optimize.brentq`) on the monotonic function $\alpha \mapsto |g_{mn}(\alpha)|^2$; Laguerre polynomials and the confluent hypergeometric function are evaluated with `scipy.special.eval_genlaguerre` and `scipy.special.hyp1f1` respectively. The complete, independently executable code — including every closed-form routine, the truncated-Fock-space validation layer, and the figure-generation routines with on-screen display and vector-PDF export — accompanies this manuscript as the Jupyter notebook `pacs_berry_uhlmann.ipynb`.

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